Chapter 1: Numerical Methods

We introduce a locally mass conserving numerical framework. We solve the static mechanics system with conforming mixed finite element method, which is adjusted to be locally mass conservative. We solve the transport system with finite volume scheme, which is a mass conservative scheme.

1.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$. Here, $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. Let $\mathcal{T}_h$ be a conforming mesh of quadrilaterals covering Ω. The quadrilateral $E \in \mathcal{T}_h$ is referred as element in finite element context, while referred as cell in finite volume context. Each quadrilateral E is assumed to be strictly convex, not degenerate into a triangle, line or point. For ease of implementation, we use a logically rectangular mesh, where each element(cell) is indexed using Cartesian coordinates $\{i, j\}$. We show a reference element(cell) $\hat{E}$ and a general quadrilateral element(cell) E in Figure 1.1. There exists a non-affine bijective and smooth map $\mathbf{F}_{\hat{E}} : \hat{E} \to E$ mapping the vertices of $\hat{E}$ to the vertices of E.

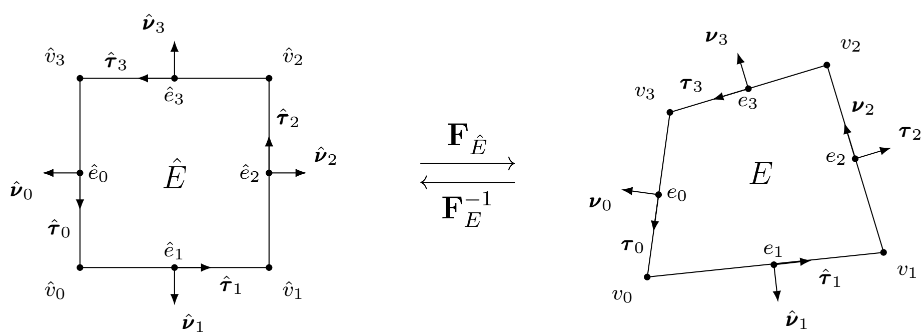

Figure 1.1: An exhibition of a reference square and a general non-degenerate quadrilateral element(cell).

The edges of element E are denoted by e_i, $i \in \{0,1,2,3\}$ in the counterclockwise direction. The vertices are represented as $v_i = e_i \cap e_{i+1}$ in the sense of modulo. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively. The unit normal vector $\boldsymbol{\nu}_i$ points outerwards the quadrilateral. The tangent $\boldsymbol{\tau}_i$ is generated by rotating $\boldsymbol{\nu}_i$ counterclockwise by right-hand rule.

In Figure 1.2, we illustrate a full logically rectangular qusi-uniform quadrilateral mesh $\mathcal{T}_h$, which is generated by randomly distort the interior vertex of a square mesh.

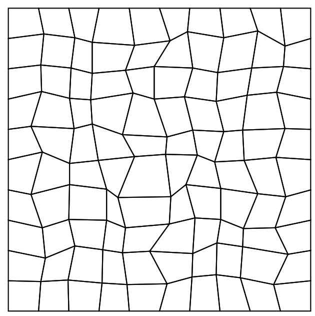

Figure 1.2: An exhibition of a full logically rectangular qusi-uniform quadrilateral mesh $\mathcal{Q}_h$.

1.1.1 Shape Regularity

We give an official definition of uniform shape regular of a quadrilateral mesh. For any quadrilateral $E \in \mathcal{T}_h$, we can get a set of four triangles $T_i, \quad i \in \{0,1,2,3\}$. We form the triangle T_i by excluding vertex v_i from the quadrilaterl E, and use the remaining three vertex. We define the following parameters as

$$\begin{aligned} h_E &= \text{diameter of} E, \\ \rho_E &= 2 \min_{1 \leq i \leq 4} \{\text{diameter of largest circle inscribed in } T_i\}. \end{aligned} \tag{1.1}$$

We consider a mesh partion $\mathcal{T}_h$ is uniformly shape regular if there exists a shape regularity parameter σ_* independent of $\mathcal{T}_h$, such that the ratio

$$\rho_E / h_E \geq \sigma_* > 0, \tag{1.2}$$

for all quadrilaterals $E \in \mathcal{T}_h$.

1.2 Direct Serendipity Space

Serendipity element have the same accuracy of approximation with classical tensor product element while use smaller number of degrees of freedom. However, the classical serendipity space suffer from poor approximation on quadrilaterals. We define local shape functions directly on quadrilaterals according to Arbogast et al. (2022). H^1 and $H(\text{div})$ conforming direct mixed finite elemenent space are constructed with the direct serendipity function space.

To begin with, we introduce some notations and distance auxilliary functions. Let $\mathbb{P}_r(\omega)$ denote the space of polynomials of degree up to r on $\omega \subset \mathbb{R}^d$. The dimension of $\mathbb{P}_r$ can be calculated as

$$\dim \mathbb{P}_r(\mathbb{R}^d) = \binom{r+2}{2} = \frac{(r+d)!}{r!d!}. \tag{1.3}$$

We define the linear polynomial $\lambda_i(\mathbf{x})$ giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to edge e_i as

$$\lambda_i(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_i^*) \cdot \boldsymbol{\nu}_i, \quad i = 0, 1, 2, 3, \tag{1.4}$$

where $\mathbf{x}_i^* \in e_i$ is any point on the edge e_i. The value of distance function λ_i keeps positive as long as $\boldsymbol{x}$ is in the interior of the quadrilateral. We define two additional distance functions giving the distance of $\mathbf{x} \in \mathbb{R}^2$ to diagonal line d_i joining v_i with v_{i+2}, $i \in 0, 1$. Let $\nu_{d,i}$ be the unit normal vector to the diagonal line d_i. The direction of the $\nu_{d,i}$ is defined by counterclockwise rotating of the vector pointing from v_i to v_{i+2}. The resulting diagonal distance function is defined as

$$\lambda_{d,i}(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_{d,i}^*) \cdot \boldsymbol{\nu}_{d,i}, \quad i = 1, 2, \tag{1.5}$$

where $\mathbf{x}_{d,i}^*$ is any point on the diagonal line d_i. For simplicity, we use the coordinate of corner v_i in later computation. We denote the coordinates of corner v_i as $\mathbf{x}_{v,i}$.

Now we present set of nodal basis functions in first and second order direct serendipity space $\mathcal{DS}_1$ and $\mathcal{DS}_2$ defined by Arbogast et al. (2022).

1.2.1 Direct Serendipity Space $\mathcal{DS}_2$

We supplement polynomial space $\mathbb{P}_r(E), r \geq 2$ with two linearly independent functions, $\phi_{s,1}(\mathbf{x})$ and $\phi_{s,2}(\mathbf{x})$. We denote the supplemental function space as $\mathbb{S}_r^{\mathcal{DS}}(E) = \text{span}\{\phi_{s,1}, \phi_{s,2}\}, \quad r \geq 2$. We present the second order direct serendipity space as

$$\mathcal{DS}_2(E) = \mathbb{P}_2(E) \oplus \mathbb{S}_2^{\mathcal{DS}}(E). \tag{1.6}$$

The supplemental functions are defined as

$$\phi_{s,i}(\mathbf{x}) = \lambda_{i+1}(\mathbf{x})\lambda_{i+2}(\mathbf{x})R_{i,i+2}(\mathbf{x}), \quad i \in \{1,2\}, \tag{1.7}$$

where i is in the sense of modulo. We use a simple definition for rational function $R_{i,i+2}$ as

$$R_{i,i+2}(\mathbf{x}) = \frac{\lambda_i(\mathbf{x}) - \lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{1.8}$$

which satisfies the conditions as

$$R_{i,i+2}|_{e_i} = -1, \quad R_{i,i+2}|_{e_{i+2}} = 1. \tag{1.9}$$

We continue to define

$$R_i(\mathbf{x}) = \frac{1}{2}\Big(1 - (-1)^{\lfloor i/2 \rfloor} R_{i,i+2}(\mathbf{x})\Big) = \frac{\lambda_{i+2}(\mathbf{x})}{\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x})}, \tag{1.10}$$

where i is understood $\pmod 2$. Here, $R_i|_{e_i} = 1$, and $R_i|_{e_{i+2}} = 0$ by definition. In Figure 1.3, we draw R_0 on a general quadrilateral, which evaluates 1 on the edge e_0, and evaluates 0 on the opposite edge e_2.

The edge nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi_{e,i}^2(\mathbf{x}) = \frac{\lambda_{i+1}(\mathbf{x})\lambda_{i+3}(\mathbf{x})R_i(\mathbf{x})}{\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.11}$$

where $\mathbf{x}_{e,i}$ is the middle point of the edge e_i, the i index is understood in the sense of modulo of 4. In Figure 1.4, we present $\varphi_{e,0}^2$ on a general quadrilateral.

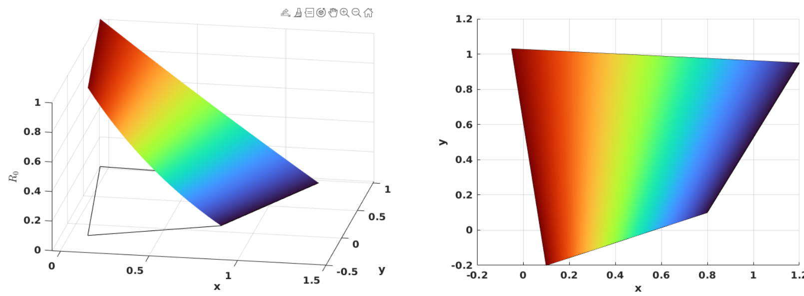

Figure 1.3: Rational function R_0.

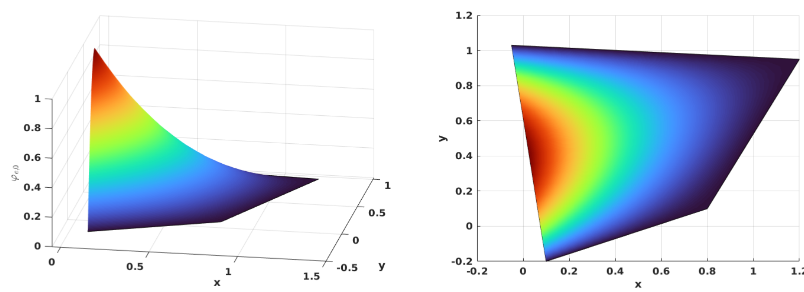

Figure 1.4: Edge nodal basis function $\varphi^2_{e,0}$ in $\mathcal{DS}_2$.

The vertex nodal basis functions for $\mathcal{DS}_2$ are constructed as

$$\varphi^2_{v,i}(\mathbf{x}) = \frac{\lambda_{i+2}(\mathbf{x})\lambda_{i+3}(\mathbf{x}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi^2_{e,i}(\mathbf{x})}{\lambda_{i+2}(\mathbf{x}_{v,i})\lambda_{i+3}(\mathbf{x}_{v,i}) - \sum_{k=i}^{i+1} \lambda_{i+2}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})\varphi^2_{e,i}(\mathbf{x}_{v,i})}, \tag{1.12}$$

where $\mathbf{x}_{v,i}$ is the coordinates of the vertex node v_i. The i index is understood in the sence of module of 4 as well. In Figure 1.5, we show $\varphi^2_{v,0}$ on a general quadrilateral.

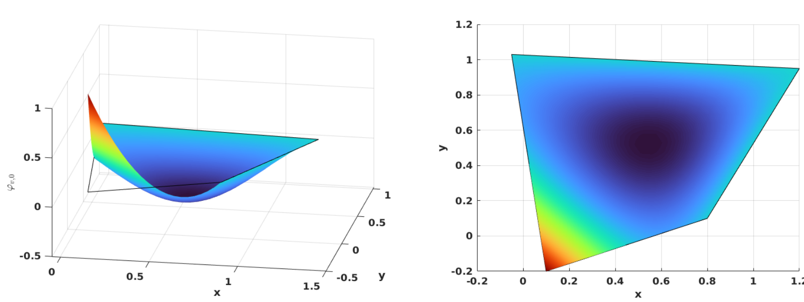

Figure 1.5: Vertex nodal basis function $\varphi^2_{v,0}$ in $\mathcal{DS}_2$.

We conclude that unisolvence follows directly from the constructed nodal basis functions.

1.2.2 Direct Serendipity Space $\mathcal{DS}_1$

The supplemental space for $\mathcal{DS}_1$ differs from that for $\mathcal{DS}_2$. We present the first order serendipity as

$$\mathcal{DS}_1(E) = \mathbb{P}_1(E) \oplus \text{span}\{R\}, \tag{1.13}$$

where $R(\mathbf{x}_{v,0}) = R(\mathbf{x}_{v,2}) = -R(\mathbf{x}_{v,1}) = -R(\mathbf{x}_{v,3}) = 1$. The construction of rational funcion R is not unique. We present one way to construct $R(\mathbf{x})$ with edge nodal basis function (1.11) in $\mathcal{DS}_2$ as

$$R(\mathbf{x}) = r(\mathbf{x}) - \sum_{i=0}^{3} r(\mathbf{x}_{e,i})\varphi^2_{e,i}(\mathbf{x}), \tag{1.14}$$

where auxiliary function $r(\mathbf{x})$ is defined as

$$r(\mathbf{x}) = \frac{\lambda_2(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_2(\mathbf{x}_{v,0})\lambda_3(\mathbf{x}_{v,0})} - \frac{\lambda_0(\mathbf{x})\lambda_3(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,1})\lambda_3(\mathbf{x}_{v,1})} + \frac{\lambda_0(\mathbf{x})\lambda_1(\mathbf{x})}{\lambda_0(\mathbf{x}_{v,2})\lambda_1(\mathbf{x}_{v,2})} - \frac{\lambda_1(\mathbf{x})\lambda_2(\mathbf{x})}{\lambda_1(\mathbf{x}_{v,3})\lambda_2(\mathbf{x}_{v,3})}. \tag{1.15}$$

In Figure 1.6, we plot the rational function on a general quadrilateral.

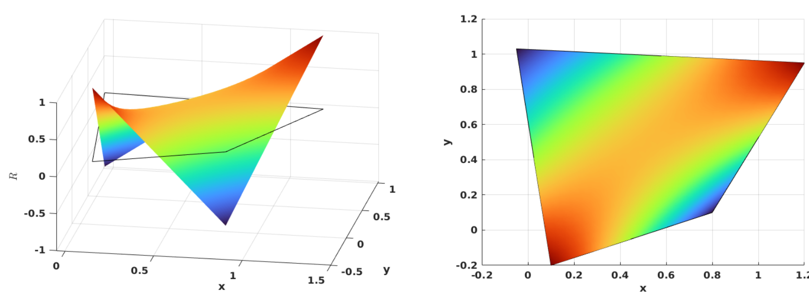

Figure 1.6: Rational function in $\mathcal{DS}_1$.

We define vertex nodal basis function accordingly as

$$\varphi^1_{v,i}(\mathbf{x}) = \frac{\lambda_{d,m}(\mathbf{x}) - \frac{1}{2}\lambda_{d,m}(\mathbf{x}_{v,i+2})(1 + (-1)^i R(\mathbf{x}))}{\lambda_{d,m}(\mathbf{x}_{v,i}) - \lambda_{d,m}(\mathbf{x}_{v,i+2})}, \tag{1.16}$$

where $m \equiv i+1 \pmod 2$, and index i is understood in the sence of modulo of 4. In Figure 1.7, we show the vertex nodal basis function $\varphi^1_{v,0}$ in $\mathcal{DS}_1$ on a general quadrilateral, which evaluates 1 at vertex v_0 and 0 at other vertex.

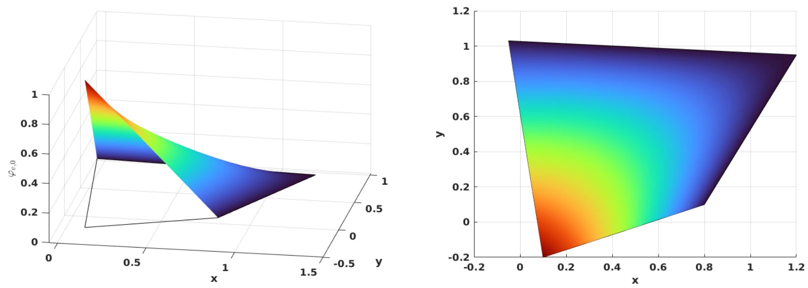

Figure 1.7: Vertex nodal basis function in $\mathcal{DS}_1$.

We first define an interpolation operator of function $v \in L^2(\Omega)$ onto $\mathcal{DS}_r(K_i)$, $r = 1, 2$. Here, K_i has different meanings with different types of basis functions. For each basis functions associated with edge node, we set $K_i = e_i$. For each basis functions associated with vertex node, we set K_i being a set of edges sharing the vertex v_i. Following the work by Arbogast et al. (2022), we define a interpolation operator $\mathcal{I}^r_h : W^l_p(E) \to \mathcal{DS}_r$ as

$$\mathcal{I}^r_h v(\mathbf{x}) = \sum_{i=1}^{N_r} \varphi_i(\mathbf{x}) \int_{K_i} \psi_i(\mathbf{y}) v(\mathbf{y}) d\mathbf{y} \in \mathcal{DS}_r, \tag{1.17}$$

where $1 \le p \le \infty$ and $l > 1/p$ (but $l \ge 1$ if $p = 1$), and $\psi(\mathbf{x})$ is the dual nodal basis such that

$$\int_{K_i} \psi_i(\mathbf{x}) \varphi_j(\mathbf{x}) d\mathbf{x} = \delta_{i,j}, \quad i, j \in \{0, 1, 2, \dots, N_r\}, \tag{1.18}$$

where N_r is the number of nodal basis defined on K_i. We present the following two lemmas regarding the constructed interpolation operator for future reference.

Lemma 1.1. *(Boundedness of the interpolation operator Arbogast et al. (2022))*

Let $\mathcal{T}_h$ be uniformly shape regular (1.2) *with shape regularity parameter σ_*. For each element $E \in \mathcal{T}_h$, There exists a bounded interpolation operator* (1.17). *For* *every $E \in \mathcal{T}_h$, suppose that the basis functions of $\mathcal{DS}_r(E)$ are constructed using λ_{24} and λ_{13} such that the zero set $\mathcal{Z}_{i,i+2}$ intersects e_{i+3} and e_{i+1}. Moreover, assume that $R_{i,i+2}$ is uniformly differentiable functions of the vertices of E up to order m. Then for $r \geq 1$, $E \in \mathcal{T}_h$, $1 \leq q \leq \infty$, and any nonnegative integer m,*

$$\|\mathcal{I}_h^r v\|_{W_q^m} \leq C(\sigma_*, m, q) \sum_{k=0}^{l} h_E^{k-m+\frac{2}{q}-\frac{2}{p}} |v|_{W_p^k(E^*)}, \tag{1.19}$$

where $E^* = \cup_{F \in \mathcal{T}_h, F \cap E \neq \emptyset}$ and $|\cdot|_{W_p^k}$ is the seminorm of k-th order derivatives.

Lemma 1.2. *(Approximability of the interpolation operator Arbogast et al. (2022))*

Under the assumptions of 1.1, there exists a constant $C = C(r, \sigma_*) > 0$ such that for all functions $v \in W_{l,p}(E^*)$, with $1 \leq p \leq \infty$ and $l > 1/p$ or ($l \geq 1$ if $p = 1$),

$$\|v - \mathcal{I}_h^r v\|_{W_p^m} \leq C h_E^{l-m} |v|_{W_p^l(E^*)}, \quad 0 \leq m \leq \min(l, r+1). \tag{1.20}$$

Morevoer, there exists a constant $C = C(r, \sigma_*) > 0$, independent of $h = \max_{E \in \mathcal{T}}$, such that for all functions $v \in W_p^l(\Omega)$,

$$(\sum_{E \in \mathcal{T}_h} \|v - \mathcal{I}_h^r v\|_{W_p^m}^p)^{1/p} \leq C h^{l-m} |v|_{W_p^l(\Omega)}, 0 \leq m \leq \min(l, r+1). \tag{1.21}$$

1.3 Mixed Finite Element

We define H^1 and $H(\text{div})$ conforming mixed element space with the direct serendipity space, and test with simple stokes and darcy problems respectively.

1.3.1 $H(\textbf{div})$ Conforming Element

We have some choice for $H(\text{div})$ confirming space. For example, we can choose the famous Ravier-Thomas element by Raviart and Thomas (1977). The serendipity space is the precursor of the Brezzi-Douglas-Marini element by Brezzi et al. (1985). We recover Arbogast-Correa element by Arbogast and Correa (2016) if we use mapped direct serendipity space. We generated a reduced $H(\text{div})$ approximation, that $\nabla \cdot \mathbf{u}$ is approximated one less order than approximation of $\mathbf{u}$.

We write the rotated 2D exact sequence for energy space as

$$H^1 \xrightarrow{\text{curl}} H(\text{div}) \xrightarrow{\text{div}} L^2 \tag{1.22}$$

$$\cup \qquad\qquad \cup \qquad\qquad \cup \tag{1.23}$$

$$\mathcal{DS}_2 \xrightarrow{\text{curl}} \mathbf{V}_r^{r-1} \xrightarrow{\text{div}} \mathbb{P}_{r-1}. \tag{1.24}$$

We define the first order reduced $H(\text{div})$ conforming space with $\mathcal{DS}_2$ as

$$\mathbb{V}_1^0 = \mathbb{P}_1^2 \oplus \text{curl}\mathbb{S}_2^{\mathcal{DS}}. \tag{1.25}$$

The reduced space $\mathbb{V}_1^0$ can be given a global basis with the normal components of the shape functions merged continuously on the shared edge of two elements. The dimension of the derived space is

$$\dim(\mathbb{V}_1^0) = 2 \times 3 + 2 = 8. \tag{1.26}$$

We need two independent basis functions on each edge of a quadrilateral E, one of which has a vanishing average flux on the edge and the other one has a non-vanishing average flux on the edge. On each edge, we define two independent basis functions. $\boldsymbol{\varphi}_{l,i}$ and $\boldsymbol{\varphi}_{c,i}$, such that

$$\boldsymbol{\varphi}_{c,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ 1, & j = i. \end{cases} \tag{1.27}$$

$$\boldsymbol{\varphi}_{l,i} \cdot \boldsymbol{\nu}_j|_{e_j} = \begin{cases} 0, & j \neq i, \\ l_i(\mathbf{x}), & j = i, \end{cases} \tag{1.28}$$

where l_i is a non vanishing linear polynomial defined on edge e_i orthogonal to one, i.e., $\int_{e_i} l_i ds = 0$.

To begin with, we write curl of the auxiliary rational function R_i (1.8) as

$$\text{curl}R_i(\mathbf{x}) = \frac{\boldsymbol{\tau}_{i+2}(\mathbf{x})\lambda_i(\mathbf{x}) - \boldsymbol{\tau}_i(\mathbf{x})\lambda_{i+2}(\mathbf{x})}{(\lambda_i(\mathbf{x}) + \lambda_{i+2}(\mathbf{x}))^2}, \tag{1.29}$$

where the index i is understood with modulo of 4. We first present a construction of basis function $\boldsymbol{\varphi}_{l,i}$ as

$$\tilde{\boldsymbol{\varphi}}_{l,i} = \text{curl}(\lambda_{i+1}\lambda_{i+3}R_i) = (\boldsymbol{\tau}_{i+1}\lambda_{i+3} + \lambda_{i+1}\boldsymbol{\tau}_{i+3})R_i + \lambda_{i+1}\lambda_{i+3}\text{curl}R_i. \tag{1.30}$$

We then normalize it such that l_i varying from -1 to 1 as

$$\boldsymbol{\varphi}_{l,i} = \frac{\tilde{\boldsymbol{\varphi}}_{l,i}(\mathbf{x})|e_i|}{4\lambda_{i+1}(\mathbf{x}_{e,i})\lambda_{i+3}(\mathbf{x}_{e,i})R_i(\mathbf{x}_{e,i})}, \tag{1.31}$$

where the index i is understood in the sense of modulo 4, and edge length $|e_i| = |\mathbf{x}_{v,i} - \mathbf{x}_{v,i+3}|$. In Figure 1.8, we draw $\boldsymbol{\varphi}_{l,2}$ on the left element and $\boldsymbol{\varphi}_{l,0}$ on the right element sharing a edge, and show normal flux joining continuously as a linear function on the shared edge.

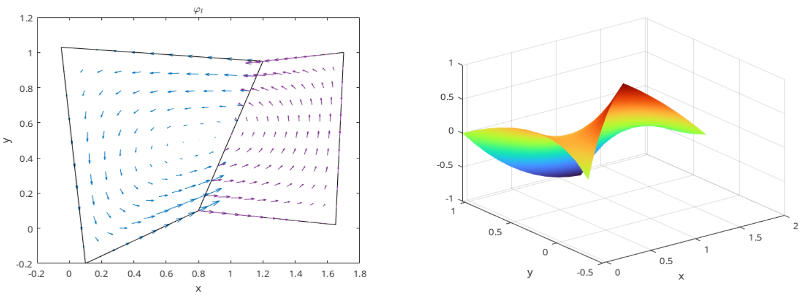

Figure 1.8: Linear basis function $\boldsymbol{\varphi}_l$.

The construction of a the constant basis function is a bit more complicated. We follow the process by Arbogast et al. (2022). We start by defining a linear nodal function $\phi^*_{v,i} \in \mathcal{DS}_2(E)$ such that $\phi^*_{v,i}(\mathbf{x}_{v,k}) = \delta_{ik}$. Its restriction to each edge of E is a linear function as

$$\phi^*_{v,i}(\mathbf{x}) = \varphi^2_{v,i}(\mathbf{x}) + \big[\frac{1}{2}\varphi^2_{e,i} + \frac{1}{2}\varphi^2_{e,i+1}\big], \tag{1.32}$$

where the index i is understood in modulo of 4. Then define $\boldsymbol{\varphi}^*_{v,i}(\mathbf{x}) = \mathrm{curl}\phi^*_{v,i}$ so that

$$\boldsymbol{\varphi}^*_{v,i} \cdot \boldsymbol{\nu}_j|_{e,j} = \begin{cases} 1/|e_i|, & j = i, \\ -1/|e_i|, & j = i+1, \\ 0, & j = i+2, i+3. \end{cases} \tag{1.33}$$

We then define the "constant" basis function defined on the edge as

$$\boldsymbol{\varphi}_{e,i} = \frac{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|\boldsymbol{\varphi}^*_{v,i} + \mathbf{x} - \mathbf{x}_{v,i+2}}{(\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_{i+1}|e_{i+1}|/|e_i| + (\mathbf{x}_{v,i} - \mathbf{x}_{v,i+2}) \cdot \boldsymbol{\nu}_i}, \tag{1.34}$$

where the index i is again understood in the sense of modulo of 4. The defined basis function $\boldsymbol{\varphi}_{e,i}$ has flux of constant value 1 on e_i and 0 on all the other edges. In Figure 1.9, we draw $\boldsymbol{\varphi}_{c,2}$ on the left element and $\boldsymbol{\varphi}_{c,0}$ on the right element sharing a edge, and show normal flux joining continuously as a constant 1 on the shared edge.

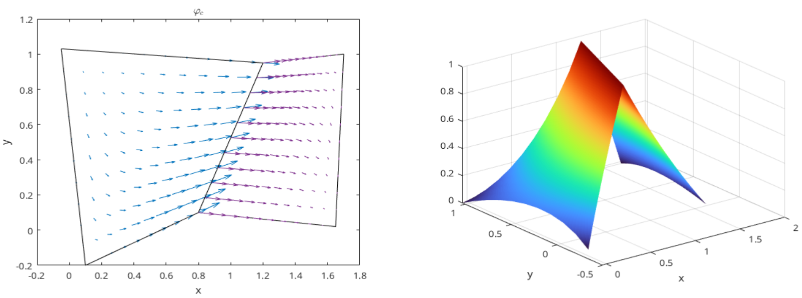

Figure 1.9: Constant basis function $\boldsymbol{\varphi}_c$.

We state the classic Ravier-Thomas global projection operator for completeness.

$$\pi_{\mathcal{RT}} : H(\text{div}; \Omega) \cap (L^{2+\epsilon}(\Omega))^2 \to \mathbb{V}_1^0, \tag{1.35}$$

where $\epsilon > 0$. We then write the approximation properties as well.

Lemma 1.3. *Stability and approximation properties Arbogast et al. (2022).*

Under the assumption of 1.1, there exists a constant $C > 0$, independent of $\mathcal{T}_h$ and $h > 0$, such that

$$\|p - p_h\|_{m,\Omega} \leq Ch^{l+1-m}|p|_{l+1,\Omega}, \quad l = 0, 1, 2, \ldots, r, \tag{1.36}$$

where $p_h \in \mathcal{DS}_2$. We have

$$\begin{aligned} \|\mathbf{u} - \pi\mathbf{u}\|_{0,\Omega} &\leq C\,\|\mathbf{u}\|_{k,\Omega}\,h^k, \qquad k = 1, \ldots, 2 \\ \|\nabla \cdot (\mathbf{u} - \pi\mathbf{u})\|_{0,\Omega} &\leq C\,\|\nabla\mathbf{u}\|_{k,\Omega}\,h^k, \quad k = 0, \ldots, 1 \\ \|p - \mathcal{P}_{W_s}p\|_{0,\Omega} &\leq C\,\|p\|_{k,\Omega}\,h^k, \qquad k = 0, \ldots, 1 \end{aligned} \tag{1.37}$$

Moreover, the discrete inf-sup condition holds

$$\sup \tag{1.38}$$

1.3.2 Test Darcy Problem

We test our $H(\text{div})$ conforming element on a simple darcy problem as

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned} \tag{1.39}$$

weak form

$$\begin{aligned} (\mathbf{u}, \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{g}, \mathbf{v})_\Omega - \langle p_0, v_n \rangle_{\Gamma_i} - \langle \mathbf{u}_0, \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned} \tag{1.40}$$

Test space $\{\mathbf{v} \in H(\text{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

1.3.3 H^1 Conforming Element

Stable mixed finite element pair for Stokes problem has been widely researched. We are referring to the element discussed by Christine Bernardi (1985) and by Fortin (1981). A more general type of element was discussed by Arbogast and Wheeler (2005).

Vertex basis mapping to sobolev space difficulty. Need Clément interpolation. Define Π operator two parts.

We consider the following space for each element $E \in \mathcal{T}_h$ as

$$V_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \text{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}, \tag{1.41}$$

where $\mathbf{b}_i$ is a quadratic bubble function defined on each edge e_i. We pick previously defined function $\varphi_{e,i} \in \mathcal{DS}_2$ as

$$\mathbf{b}_i = \varphi_{e,i}(\mathbf{x})\mathbf{n}_i \tag{1.42}$$

which is quadratic when restricted to edge e_i.

Lemma 1.4. *For an element $E \in \mathcal{T}_h$, $\mathbf{u}_h$ is uniquely defined by the following degrees of freedom:*

(i) for corner point v_i and Cartesian direction unitnormal vectors $\{\mathbf{i}, \mathbf{j}\}$,

$$DOF^{(i)}_{(v_i,\mathbf{i})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{i}, \quad DOF^{(i)}_{(v_i,\mathbf{j})}(\mathbf{u}_h) = \mathbf{u}_h(v_i) \cdot \mathbf{j};$$

(ii) for edge e_i,

$$DOF_{e_i}^{(ii)} = \langle \mathbf{u}_h \cdot \boldsymbol{\nu}_i, 1 \rangle,$$

Proof. We prove the lemma by construction. We pick nodal badis on vertex v_i in two Cartesian direction as

$$N_{v_i,x} = \begin{bmatrix} \psi_{v,i}(\mathbf{x}) \\ 0 \end{bmatrix}, \quad N_{v_i,y} = \begin{bmatrix} 0 \\ \psi_{v,i}(\mathbf{x}) \end{bmatrix} \tag{1.43}$$

We pick edge basis on edge e_i as

$$N_e = \varphi_{e,i}\boldsymbol{\nu}_i \tag{1.44}$$

Note that bubble functions are quadratic when restricted to edge e_i, while nodal basis functions are linear when restricted to edge e_i, therefore bubble functions cannot be in the same space as nodal basis function. We conclude that this element is well-defined, i.e., the degrees of freedom uniquely determine the function. □

We need a π operator, which mappes $(H^1(E))^2$ onto $V_1^{\mathcal{BR}}$ for our spaces. For $\mathbf{v} \in (H^1(\Omega))^2$, let $\pi\mathbf{v} \in \mathbb{V}_1^{\mathcal{BR}}$ be defined as the interpolant of the degrees of freedom from 1.4. For each F, we require the following as

(i) For each corner point $v \in \partial F$

$$\pi\mathbf{v}(\mathbf{x}_v) \cdot \boldsymbol{i} = \mathcal{I}_{\mathcal{DS}_1}\mathbf{v}(\mathbf{x}_v) \cdot \boldsymbol{j}. \tag{1.45}$$

(ii) On each edge $e \subset \partial E$,

$$\langle \pi\mathbf{u} \cdot \boldsymbol{\nu}, \lambda_0 \rangle = \langle \mathbf{u} \cdot \boldsymbol{\nu}, \lambda_0 \rangle_e \ \ \forall \lambda_0 \in P_0(e). \tag{1.46}$$

The interpolation operator $I_{\mathcal{DS}_1}$ has been defined in 1.17. We claim that we can represent for $\mathbf{u} \in (H^1(\Omega))^2$,

$$\pi\mathbf{u} = \mathbf{I}_{\mathcal{DS}_1}\mathbf{u} + \sum_{i=0}^{3} \alpha_i \mathbf{b}_i, \tag{1.47}$$

where the coefficient $\alpha_i = \left(\int_{e_i}(\mathbf{u} - \mathcal{I}_{\mathcal{DS}_1}\mathbf{u}) \cdot \boldsymbol{\nu}_i d\sigma\right) / \int_{e_i} \mathbf{b}_i \cdot \boldsymbol{\nu}_i d\sigma$. It is obvious that $\pi\mathbf{u} \in \mathbb{V}_1^{\mathcal{BR}}$. We now prove the approximability and stability of our projection operator and eventually derive the $\inf - \sup$ condition.

Lemma 1.5. *Assuming that $u \in (H1(\Omega))^2$.*

(i) there exists a bounded linear operator π

(ii) $\forall \mathbf{v} \in (H^1(\Omega))^2$

1.3.4 Test Stokes Problem

We test our modified Bernardi-Raugel elements on the incompressible isotropic elasticity problem with Dirichlet boundary condition:

Find the displacement $\mathbf{u}$ and pressure p such that

$$\begin{aligned} -\nabla \cdot \boldsymbol{\tau}(\mathbf{u}) + \nabla q &= \mathbf{f} \quad \text{in } \Omega, \\ \nabla \cdot \mathbf{u} &= 0 \quad \text{in } \Omega, \\ \mathbf{u} &= \mathbf{g} \quad \text{on } \partial\Omega, \end{aligned} \tag{1.48}$$

where $f \in L^2(\Omega)$, and $\boldsymbol{\tau}(\mathbf{u}) = 2(\mathcal{D}\mathbf{u} - 1/3\nabla \cdot \mathbf{u}\mathbf{I})$. The symmetrical gradient $\mathcal{D}\mathbf{u} = 1/2(\nabla\mathbf{u} + \nabla\mathbf{u}^T)$.

Define a simple choice of energy space

$$\begin{aligned} \mathbb{V} &= \{\mathbf{v} \in (H^1\Omega)^2 : \mathbf{v} = \mathbf{0} \quad \text{on } \partial\Omega\}, \\ \mathbb{U} &= \{\mathbf{u} \in (H^1\Omega)^2 : \mathbf{u} = \mathbf{g} \quad \text{on } \partial\Omega\} = \tilde{\mathbf{g}} + \mathbb{V}, \end{aligned} \tag{1.49}$$

where $\tilde{\mathbf{g}}$ is a finite energy lift. We assume that $\mathbf{g} \in (H^{1/2}(\Omega))^d$ and understand as a continuous extension $\tilde{\mathbf{g}} \in (H^1(\Omega))^d$ from the boundary into the domain. We write the problem in weak form as:

Find $\mathbf{u} \in \mathbb{U}$ and $p \in L^2(\Omega)$ such that

$$\begin{aligned} (\boldsymbol{\tau}, \nabla\mathbf{v}) - (p, \nabla \cdot \mathbf{v}) &= (\mathbf{f}, \mathbf{v}), \quad \forall \mathbf{v} \in \mathbb{V}, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \qquad \forall \omega \in L^2(\Omega). \end{aligned} \tag{1.50}$$

We consider the test problem (1.50) defined on a square region $\Omega = [-1, 1]^2$. We manufacture a displacement solution $\mathbf{u} = [x^3y^2, -x^2y^3]$ and pressure $p = y - x$. The source term can be calculated as

$$\mathbf{f} = -2 \begin{bmatrix} 3xy^2 + x^3 \\ -3x^2y - y^3 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \end{bmatrix}. \tag{1.51}$$

Solution are computed on rectangular and quadrilateral meshes. The convergence rate versus the number of elements will be presented.

1.4 Coupled Degenerate System

Consider a degenerate scaled darcy-stokes coupled boundary-value problem:

Find velocity-pressure potential pair $(\check{\mathbf{v}}_r, \check{q}_l)$ and $(\check{\mathbf{v}}_s, \check{q})$ such that

$$\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{1.52}$$

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.53}$$

$$\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{1.54}$$

$$\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.55}$$

and

$$\begin{aligned} \check{\mathbf{v}}_r &= \mathbf{g}_r \quad \text{on } \Gamma_{u,D}, \\ \check{\mathbf{v}}_s &= \mathbf{g}_s \quad \text{on } \Gamma_{u,S}, \\ p &= p_0 \quad \text{on } \Gamma_{i,D}, \\ \boldsymbol{\tau}\cdot\boldsymbol{\nu} - p &= \boldsymbol{t}_0 \quad \text{on } \Gamma_{i,S}, \end{aligned} \tag{1.56}$$

where $\Gamma_{u,D}$ and $\Gamma_{u,S}$ denote the part of boundary we assign essential boundary conditions for Darcy and Stokes problem respectively, and $\Gamma_{i,D}$ and $\Gamma_{i,S}$ represent the part of boundary we assign natural boundary conditions respectively. We have $\Gamma_{u,D} \cup \Gamma_{i,D} = \partial\Omega$ and $\Gamma_{u,S} \cup \Gamma_{i,S} = \partial\Omega$.

We follow the procedure by Arbogast et al. (2017) to discretize this boundary value problem. To begin weith, a new Hilbert space is defined for this specific scaled formulation, where the scaled velocity $\check{\mathbf{v}}_r$ should lie in, so

$$\mathbb{V}_d^{\phi} = H_{\phi}(\mathrm{div};\Omega) = \Big\{ \mathbf{v} \in (L^2(\Omega))^2 : \phi^{-1/2}\nabla \cdot (\phi^{1+\Phi}\mathbf{v}) \in L^2(\Omega) \Big\}, \tag{1.57}$$

which equipped with the inner product as

$$(\mathbf{u}_{\phi}, \mathbf{v}_{\phi})_{\mathbb{V}_r^{\phi}} = (\mathbf{u}_{\phi}, \mathbf{v}_{\phi}) + (\phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\mathbf{u}_{\phi}), \phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\mathbf{v}_{\phi})). \tag{1.58}$$

The normal trace on $\partial\Omega$ is well-defined as well as

$$\left\| \phi^{1/2+\Theta}\mathbf{v}_{\phi} \cdot \nu \right\|_{-1/2,\partial\Omega} < C_{\Omega}\Big\{ \left\|\mathbf{v}_{\phi}\right\| + \left\| \phi^{-1/2}\nabla \cdot (\phi^{1+\Theta}\mathbf{v}_{\phi}) \right\| \Big\}. \tag{1.59}$$

1.4.1 The weak formulation

We define the following function spaces

$$\begin{aligned}
\mathbb{V}_D^{\phi} &= \Big\{ \mathbf{v} \in H_{\phi}(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu} = 0 \text{ on } \Gamma_{u,D} \Big\}, \\
\mathbb{W}_D &= L^2(\Omega), \\
\mathbb{V}_S &= \Big\{ \mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S} \Big\}, \\
\mathbb{W}_S &= L^2(\Omega),
\end{aligned} \tag{1.60}$$

where each space is equipped with their natural norm. To empose essential boundary conditoins, we assume that a function $\mathbf{g}_s = \check{\mathbf{v}}_{s,0}$ on $\Gamma_{u,S}$ and $\mathbf{g}_s = \mathbf{0}$ on $\partial\Omega/\Gamma_{u,S}$. Then $\mathbf{g}_s \in (H^{1/2}(\partial\Omega))^2$ and extended it continuously from the boundary to the domain, so that the extension $\mathbf{g}_s \in (H^1(\Gamma_{u,S}))^2$ and $\left\|\mathbf{g}_{s,0}\right\|_1 \leq C \left\|\mathbf{v}_{s,0}\right\|_{1/2,\Gamma_{u,S}}$. Similarly, we can define a bounded extension of $\check{\mathbf{v}}_r \in \mathbb{V}_D^{\phi}$ such that

Scaled nondimensionalized weak form. Find $\check{\mathbf{v}}_r \in \mathbb{V}_D^{\phi} + \tilde{\mathbf{g}}_r$, $\check{q}_l \in \mathbb{W}_D$, $\check{\mathbf{v}}_s \in \mathbb{V}_S + \tilde{\mathbf{g}}_S$, and $\check{q} \in \mathbb{W}_S$ such that

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_s), \nabla\boldsymbol{\psi}_s) - (\check{q}, \nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall\boldsymbol{\psi}_s \in \mathbb{V}_S, \tag{1.61}$$

$$(\nabla\cdot\check{\mathbf{v}}_s, \omega) + \left(\frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega\right) = 0 \qquad \forall\omega \in \mathbb{W}_S, \tag{1.62}$$

$$(\check{\mathbf{v}}_r, \boldsymbol{\psi}_r) - (\check{q}_l, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r)) = 0, \qquad \forall\boldsymbol{\psi}_r \in \mathbb{V}_D^{\phi}, \tag{1.63}$$

$$(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \left(\frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega_f\right) = 0 \qquad \forall\omega_f \in \mathbb{W}_D. \tag{1.64}$$

The Existence and uniquense of the solution has been proved by Arbogast et al. (2017). We will not repeat the proof here.

1.4.2 MFEM discretization

We discretize the formed degenerate system with previously defined $H(\text{div})$ and H^1 conforming space. We write the function spaces out as

$$\begin{aligned}
\mathbb{V}_1^0 &= (\mathbb{P}_1)^2 \oplus \text{curl}\mathbb{S}_2^{\mathcal{DS}} \\
\mathbb{W} &= \mathbb{P}_0 \\
\mathbb{V}_1^{\mathcal{BR}} &= \mathcal{DS}_1 \oplus \text{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \\
\mathbb{W} &= \mathbb{P}_0.
\end{aligned} \tag{1.65}$$

We can impose essential boundary conditions

Scaled mixed finite element method. Find $\check{\mathbf{v}}_{r,h} \in \mathbb{V}_1^0 + \hat{\mathbf{g}}_r$, $\check{q})_{r,h} \in \mathbb{W}_D$, $\mathbf{v}_{s,h} \in \mathbf{V}_S + \hat{\mathbf{g}}_s$ and $\check{q}_h \in \mathbb{W}_S$ such that

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_{s,h}), \nabla\boldsymbol{\psi}_s) - (\check{q}_h, \nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall\boldsymbol{\psi}_s \in \mathbb{V}_1^{\mathcal{BR}}, \tag{1.66}$$

$$(\nabla\cdot\check{\mathbf{v}}_{s,h}, \omega) - \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega\Big) = 0 \qquad \forall\omega \in \mathbb{W}_0^{\mathcal{BR}}, \tag{1.67}$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_r) - (\check{q}_{l,h}, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall\boldsymbol{\psi}_r \in \mathbb{V}_1^0, \tag{1.68}$$

$$(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}), \omega_f) + \Big(\frac{1}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_f\Big) = 0 \qquad \forall\omega_f \in \mathbb{W}_0. \tag{1.69}$$

1.4.3 Local Mass Conservation

Before we continue the discussion, we need to make sure the formulation is well-defined by not dividing by zero. The term $\phi^{-1/2}$ in the symmetric equations (1.68) and (1.69) causes trouble when there is no melting, i.e., $\phi = 0$. In order to deal with it, we first define a element-averaged $\overline{\phi}_E$ as

$$\overline{\phi}_E = \frac{1}{|E|}\int_E \phi(\mathbf{x})d\mathbf{x}, \tag{1.70}$$

where $|E|$ is the area of element E. Let

$$\phi_E = \begin{cases} \overline{\phi}_E & \text{if } \overline{\phi}_E \neq 0, \\ 0 & \text{if } \overline{\phi}_E = 0. \end{cases} \tag{1.71}$$

We now replace the $\phi^{-1/2}$ with $\phi_E^{-1/2}$. Then we expand the integral with divergence theorem as

$$\begin{aligned} \int_E \phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_{r,h}) \omega_l = &\int_{\partial E} \phi_E^{-1/2} \phi^{1+\Theta} \boldsymbol{\psi}_{r,h} \cdot \boldsymbol{\nu} \omega_f ds \\ &- \int_E \phi_E^{-1/2} \phi^{1+\Theta} \boldsymbol{\psi}_{r,h} \cdot \nabla \omega_l d\mathbf{x}. \end{aligned} \tag{1.72}$$

The second term vanishes when ω_l is constant on element E. We define an auxiliary auxiliary space ω_E begin 1 on E and 0 elsewhere. We sum equations (1.67) and (1.69) together assuming that $\omega = \omega_E \in \mathbb{W}_S$ and $\omega_l = \phi_E^{-1/2} \omega_E \in \mathbb{W}_D$. We can achieve local mass conservation of the phase averaged mixture if we replace the term $\phi^{1/2}$ in equation (1.67) with $\phi_E^{1/2}$ as

$$\int_E \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_r + \check{\mathbf{v}}_s) d\mathbf{x} = 0. \tag{1.73}$$

We now propose our local mass conserved version as

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_{s,h}), \nabla \boldsymbol{\psi}_s) - (\check{q}_h, \nabla \cdot \boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall \boldsymbol{\psi}_s \in \mathbb{V}_1^{\mathcal{BR}}, \tag{1.74}$$

$$(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega) - \left(\frac{\phi_E^{1/2}}{1-\phi} (\check{q}_{l,h} - \phi^{1/2} \check{q}_h), \omega \right) = 0 \qquad \forall \omega \in \mathbb{W}_0^{\mathcal{BR}}, \tag{1.75}$$

$$(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_r) - (\check{q}_{l,h}, \phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \boldsymbol{\psi}_{r,h})) = 0, \qquad \forall \boldsymbol{\psi}_r \in \mathbb{V}_1^0, \tag{1.76}$$

$$(\phi_E^{-1/2} \nabla \cdot (\phi^{1+\Theta} \check{\mathbf{v}}_{r,h}), \omega_f) + \left(\frac{1}{1-\phi} (\check{q}_{l,h} - \phi^{1/2} \check{q}_h), \omega_f \right) = 0 \qquad \forall \omega_f \in \mathbb{W}_0. \tag{1.77}$$

1.4.4 Convergence Rate

1.4.5 Test Run

We test our MFEM setup with two problems. First, we test with a manufactured solution. Second, we test a more complex yet a more realistic problem.

Problem setup 1:

We solve a manufactured system of equations of the rectangular domain $-1 < x < 1$,

$$\mathbf{v}_s = \tag{1.78}$$

Problem setup 2:

We solve the full system of equations on the rectangular domain $-160\text{km} < x < 160\text{km}$, $0, x, 160\text{km}$. The boundary conditions are define by imposing

$$\mathbf{v}_s = \tag{1.79}$$

1.5 Saddle Point System

Saddle point system is widely spread in technical and science scenarios, such as constrained optimization problem and has been widely researched. In our case, saddle point system is formed along with Mixed Finite Element discretization. In this section, we give a brief overview of characteristics of a saddle point system and some common way to solve such system. A saddle point system is formed as

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{g} \end{bmatrix}. \tag{1.80}$$

We made the following assumptions

- $\mathbf{A}$ is symmetric: $\mathbf{A} = \mathbf{A}^T$;
- $\mathbf{A}$ is nonsingular: $\mathbf{A}^{-1}$ exists;
- $\mathbf{C}$ is symmetric and positive semidefinite;
- $\mathbf{B}$ is rank deficient.

The null space of $\mathbf{B}$ matrix $\ker(\mathbf{B})$ in our darcy problem is not fixed due to evloution of melting (change of porosity ϕ). Considering the need of parallel, a proper iterative method is discussed.

1.5.1 SPS Induced by MFEM

Now we discuss the saddle point system (SPS) induced by MFEM method in details including the boundary conditions. To begin with, we regroup dofs according to the boundary condition.

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_K & \mathbf{A}_M \\ \mathbf{A}_M^T & \mathbf{M} \end{bmatrix}, \tag{1.81}$$

where $\mathbf{A}_K$ consists of interaction within interior dofs and natural boundary condition dofs, and $\mathbf{A}_M$ consists of iteraction between interior dofs, natural boundary dofs and essential boundary dofs.

1.5.2 Degenerate Coupled Saddle Point Problem and Uzawa Iteration

The linear system formed from the equations (1.74), (1.75), (1.76) and (1.77) is

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}, \tag{1.82}$$

wherein the solution represents the degrees of freem of primary variables with respect to the bases of the finite element spaces. (Insert details of calculation of each blocked matrix). We form a double saddle point system by symmetrically permutate original linear system,

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}. \tag{1.83}$$

We group these block matrix to form a concise version,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}, \tag{1.84}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{1.85}$$

where $\mathbf{A}$ is a positive definite matrix. However, $\mathbf{B}$ is usually a singular matrix when no melting region presents. We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system. Uzawa algorithm applied to the case of discretizing Stokes problem with Mixed Finite Element has beed discussed by Elman and Golub (1994) and Bramble et al. (1997),

Algorithm 1 Coupled Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
3: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
4: Obtain $\tilde{\mathbf{y}} = \text{argmin}_{\mathbf{x}\in V}||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$
5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

1.6 Finite Volume Method

The finite volume method (FVM) is a discretization method widely used in various scenarios that conservation laws are involved. The finite volume is a robust algorithm that can be applied to arbitriary complex geometry. The most important aspect of FVM that benefits this simulation is that the local mass conservation is achieved naturally. To begin with, we consider a general conservation law retarding a scalar variable u as

$$\begin{cases} \dfrac{\partial u(\mathbf{x},t)}{\partial t} + \nabla\cdot\mathbf{f}(u;\mathbf{x},t) = 0, & t>0, \mathbf{x}\in\Omega, \\ u(\mathbf{x},0) = u_0(\mathbf{x}), & \mathbf{x}\in\Omega, \end{cases} \tag{1.86}$$

where we assume the domain $\Omega\subset\mathbb{R}^2$. The term $\mathbf{f}$ is a flux transporting scalar variable u. We write the basic semi-discretized form of basic finite volume method as

$$\frac{\partial\overline{u}}{\partial t} + \frac{1}{|E|}\int_{\partial E}\mathbf{F}(\mathbf{x},t)\cdot\boldsymbol{\nu}d\mathbf{x} = 0, \tag{1.87}$$

where we use the cell-averaged value $\overline{u} = \frac{1}{|E|}\int_E u(\mathbf{x},t)d\mathbf{x}$, and $F(\mathbf{x},t)$ is the approximated numerical flux. We use the Lax-Friedrichs numerical flux

$$\mathbf{F}(u^+,u^-) = \frac{1}{2}[(\mathbf{f}(u^+)+\mathbf{f}(u^-))\cdot\boldsymbol{\nu} - \alpha_{LF}(u^+-u^-)], \tag{1.88}$$

where u^+ and u^- represent values approximated from element $E^+ \in \mathcal{T}_h$ and $E^- \in \mathcal{T}_h$ respectively, and $\boldsymbol{\nu}$ is the unit vector pointing from E^+ to E^- orthogonal to the edge. Here, α_{LF} is the Lax-Fredrich stabilizer factor. We have two different ways to compute α_{LF} depending on the knowledge of solution.

- The global LF Stabilizer: $\alpha_{LF} = \max_u |\partial \mathbf{f} \cdot \boldsymbol{\nu} / \partial u|$;
- The local LF Stabilizer: $\alpha_{LF} = \max(|\partial \mathbf{f}^+ \cdot \boldsymbol{\nu} / \partial u^+|, |\partial \mathbf{f}^- \cdot \boldsymbol{\nu} / \partial u^-|)$.

The numerical flux $\mathbf{F}$ (1.88) is consistent and conservtive across the edge. We now replace the u^+ and u^- with reconstructed values and rewrite the numerical flux as

$$\mathbf{F}(\mathcal{R}^+, \mathcal{R}-) = \frac{1}{2}[(\mathbf{f}(\mathcal{R}^+) + \mathbf{f}(\mathcal{R}^-) \cdot \boldsymbol{\nu} - \alpha_{LF}(\mathcal{R}^+ - \mathcal{R}^-)], \tag{1.89}$$

where $\mathcal{R}$ is a higher order reconstruction value.

1.7 The Multi-level Weighted Essentially Non-Oscillatory (ML-WENO) Method

We develop a accurate, versatile, robust and efficient finite volume weighted essentially non oscillatory (WENO) techniques for reconstruction of values on both sides of the edge in Arbogast et al. (2024).

1.7.1 Stencil Polynomial

Given a stencil $\mathcal{S} = \{E_0, E_2, \ldots, E_k\}$ of the quasiuniform logically rectangular mesh. We can define an associated tensor product stencil polynomial $Q_{s,t} \in \mathbb{Q}_{(s,t)} := \mathbb{P}_{s-1}(x) \otimes \mathbb{P}_{t-1}(y)$, where s and t equal to the stencil sizes in the x- and y-directions, respectively. We scale our stencil polynomial individually as

$$Q_{s,t} = \sum_{p<s,q<t} c_{p,q} \Big(\frac{x - x_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^p \Big(\frac{y - y_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^q, \tag{1.90}$$

where $(x_{\mathcal{S}}, y_{\mathcal{S}})$ is the center point of the stencil $\Omega_{\mathcal{S}} = \cup_{E\in\mathcal{S}} \subset \mathbb{R}^2$, and $h_{\mathcal{S}} = \sqrt{\sum_k E_k/k}$ is the diameter of average cell size. Let $\xi = (x - x_{\mathcal{S}})/h_{\mathcal{S}}$ and $\eta = (y - y_{\mathcal{S}})/h_{\mathcal{S}}$ be the normalized varaibles, we can rewrite it into normalized version:

$$\hat{Q}_{s,t} = \sum_{p<s,q<t} c_{p,q}\xi^p\eta^q. \tag{1.91}$$

In finite volume scenerio, we are dealing with cell averaged value of the scalar transport variable

$$\overline{u}_E = \frac{1}{|E|}\int_E u(\mathbf{x})d\mathbf{x} = \frac{1}{|E|}\int_E u(x,y)dxdy. \tag{1.92}$$

We can calculate coefficents $c_{p,q}$ by imposing a constraint that cell averaged value (1.92) is preserved through integral with stencil polynomial

$$\frac{1}{|E_k|}\int_{E_k} Q(\mathbf{x})d\mathbf{x} = \overline{u}_{E_k}, \quad \forall E_k \in \mathcal{S}. \tag{1.93}$$

We will replace $\overline{u}_{E_k}$ with $\overline{u}_k$ in the following discussion for simplicity.

We can represent stencil polynomial with basis polynomials $Q_{s,t} = \sum_k \overline{u}_k \phi^k_{s,t}$, where basis polynomials can be calculated as

$$\frac{1}{|E_{k'}|}\int_{E_{k'}} \phi^k_{s,t}(\mathbf{x})d\mathbf{x} = \frac{1}{|\hat{E}_{k'}|}\iint_{\hat{E}_{k'}} \hat{\phi}^k_{s,t}(\xi,\eta)d\xi d\eta = \begin{cases} 1 & k' = k \\ 0 & k' \neq k \end{cases} \quad \forall (E_{k'}, E_k) \in \mathcal{S}, \tag{1.94}$$

which would lead to a $(s\times t)\times(s\times t)$ non-singular linear system. We can now represent stencil polynomial with normalized basis polynomials as:

$$\hat{Q}_{s,t}(\xi,\eta) = \sum_k \overline{u}_k \hat{\phi}^k_{s,t}(\xi,\eta). \tag{1.95}$$

We define a linear projection operator $\pi_{\mathcal{S}}$ and let $\pi_{\mathcal{S}} u_{\mathcal{S}} = Q$ be a linear projection onto polynomial space W_p such that $\mathbb{Q}_{s,t} \subset W_p$. We estimate interpolation error with the Bramble-Hilbert Lemma Dupont and Scott (1980) and Bramble and Hilbert (1970) in $H_r(\Omega)$ space, where $r = s$ in x direction and $r = t$ in y-direction.

Lemma 1.6. *Let Ω_S be the domain covered by stencil $\mathcal{S}$ in $\mathbb{R}^2$ and let $u \in H^r(\Omega_{\mathcal{S}})$. We have:*

$$\|u - \pi_S u\|^2 \leq h^{2(r-1)} |u|^2_{H^r(\Omega_S)}. \tag{1.96}$$

Proof.

$$\|u - \pi_S u\| = \tag{1.97}$$

□

1.7.2 Stencil polynomial smoothness indicators

We define a smoothness indicator σ as a measurement of smoothness of $u(\mathbf{x})$ on the stencil S. The classic method from Jiang and Shu (1996) is defined by Sobolev seminorm as:

$$\sigma_P = \sum_{1\leq|\alpha|\leq m} \int\int_{E_0} |E_0|^{|\alpha|-1} (\mathcal{D}^\alpha Q(x,y))^2 dxdy, \tag{1.98}$$

where $\alpha = (\alpha_1, \alpha_2)$ is a multi-index, $|\alpha| = \alpha_1 + \alpha_2$, and $\mathcal{D}$ is the derivative operator $\mathcal{D}^\alpha = \partial^{|\alpha|}/\partial x^{\alpha_1} \partial y^{\alpha_2}$. The Jiang-Shu smoothness indicator can be reformed with basis polynomials:

$$\begin{aligned}
\sigma_P &= \sum_{1\leq|\alpha|\leq m} |E_0|^{|\alpha|-1} \iint_{E_0} (\mathcal{D}^\alpha Q(x,y))^2 dxdy \\
&= \sum_{1\leq|\alpha|\leq m} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi,\eta))^2 d\xi d\eta \\
&= \sum_{1\leq|\alpha|\leq m} \iint_{\hat{E}_0} \Big(\sum_{ij} \overline{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi,\eta)\Big)^2 d\xi d\eta \\
&= \sum_{ij} \sum_{i'j'} \overline{u}_{ij} \overline{u}_{i'.j'} \sum_{1\leq|\alpha|\leq m} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi,\eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi,\eta) d\xi d\eta \\
&= \sum_{ij} \sum_{i'j'} \overline{u}_{ij} \overline{u}_{i'j'} \sigma_{ij}^{i'j'},
\end{aligned} \tag{1.99}$$

where $\sigma_{ij}^{i'j'}$ can be precomputed after mesh is established after mesh is established.

We proposed an alternative way of defining smoothness indicator in Huang et al. (2023).

$$\sigma_{\text{simp}} = \frac{1}{N_{\mathcal{S}} - 1} \sum_k (\overline{u}_k - \overline{u}_0)^2, \tag{1.100}$$

where $\overline{u}_0$ is the cell averaged solution in target cell. We define this simple smoothness indicator in order to reduce the compuational cost and to deal with distorted stencil on unstructured mesh. However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator. Hence, we stick to the classic definition in the following discussion.

We observe that the smoothness indicator exhibit following convergence property that there exists some value $C > 0$ such that when $h \to 0^+$,

$$\sigma = \begin{cases} Ch^2 + \mathcal{O}(h^3) & \text{Smooth,} \\ \Theta(1) & \text{Not Smooth,} \end{cases} \tag{1.101}$$

where stencils are defined on quasi uniform mesh. This observation comes directly from the scaling property of Sobolev seminorm.

1.7.3 ML-WENO weighting

Given a target cell $E_{i,j}$, we can create a nonempty set of stencils $\{\mathcal{S}_l \ni E_{i,j}, l = 0, 1, \ldots, L, L \geq 0\}$. The reconstruction of u on target cell $E_{i,j}$ is a convex combination of stencil polynomials (1.95)

$$\mathcal{R}(\mathbf{x}) = \sum_l \tilde{\omega}_l Q_l(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \tag{1.102}$$

where $\sum_l \tilde{\omega}_l = 1$ is normalized nonlinear weights associated with stencils accordingly. We begin by arbitrarily chosing a set of linear weights $\{\omega_l\}$ and scale with stencil smoothness indicator as

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \tag{1.103}$$

where $r_l = \max\{s, t\}$, $\epsilon = 1e-4$ and $a > 0.5$ are picked as constants. The biasing factor $n_l = n(r_l)$ is used to biasing away further from constant level, whose smoothness

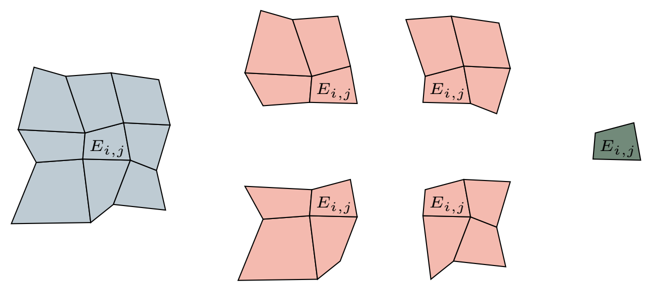

Figure 1.10: An example of (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh.

indicator is always zero, when the solution is smooth. We pick

$$n(r_l) = n_l = \begin{cases} 1, & \text{if} \quad r_l = 1, \\ 3, & \text{if} \quad r_l = 2, \\ 4, & \text{if} \quad r_l \geq 3. \end{cases} \tag{1.104}$$

We then renormalize scaled weights

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \tag{1.105}$$

1.7.4 The ML-WENO Reconstruction Error

In this section, we consider the accuracy of our multi-level weighting scheme. We group the set of indices $l \in \mathcal{L} = \{0, 1, 2, \ldots L\}$ according to the order of reconstruction polynomials, i.e., the shape of individual stencil. We denote the set of indices of stencils have s cells in x direction and t cells in y direction as $\mathcal{L}_{s,t}$. We make the assumption that the mesh resolution is fine enough so that there exist at least one stencil that is smooth if shock presents. We define two sets holding indices of stencils with or without jump as

$$\begin{aligned} \text{Smooth} &= \{l : \;\; u \text{ is smooth on } \mathcal{S}_l, \} \\ \text{Jump} &= \{l : \;\; u \text{ has a jump on } \mathcal{S}_l\} \end{aligned} \tag{1.106}$$

where we assume Smooth $\cup$ Jump $= \mathcal{L}$ and Smooth $\cap$ Jump $= \emptyset$. We represent the set of indices of stencils of shape (s, t) that is smooth as $\mathcal{L}_{s,t} \cap$ Smooth, and the set of stencils of shat (s,t) that has a jump as $\mathcal{L}_{s,t} \cap \text{Jump}$. For convience, we assume square stencils, where $r = s = t$. The accuracy result can be applied to $s \neq t$ situation directly in the selected direction. We show that the reconstruction is accurate with respect to the target cell E_0 to order

$$r_{\max} = \max\{r : \text{Smooth} \cap \mathcal{L}_{r,r} \neq \emptyset\}. \tag{1.107}$$

In Figure 1.11, we illustrate a (3,2,1) combination of stencils with thow shock presents. Here, Smooth $= \{0,1\}$, which indicates that stencil $\mathcal{S}_0$ and $\mathcal{S}_1$ do not have jump present. In this case, we expect the accuracy of reconstruction is $r_{\max} = \text{Smooth} \cap \mathcal{L}_{2,2} = 2$.

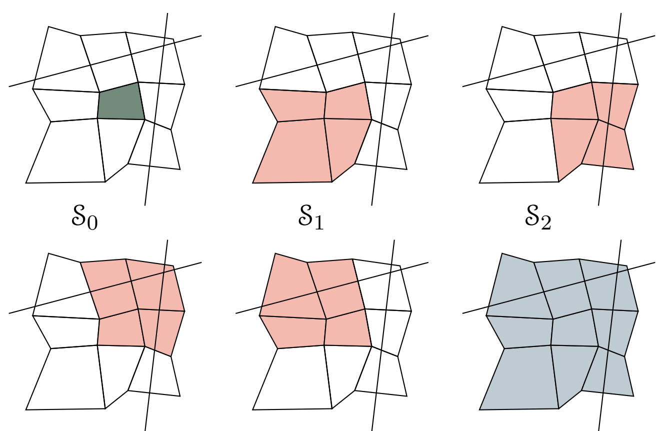

Figure 1.11: An illustration of a (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh with two shock presents.

We state the estimation of reconstruction error show in Arbogast et al. (2024). We start by estimating the error of our nonlinear scaling (1.103) asscoiated to some stencil $\mathcal{S}_l$ for some $m > 0$.

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m}. \tag{1.108}$$

We estimate the reconstruction error for two situtaions.

- $l \in \text{Smooth}$, the smoothness indicator $\sigma_l = Ch^2 + \mathcal{O}(h^3)$;

- $l \in \text{Jump}$, the smoothness indicator $\sigma_l = \Theta(1)$.

Use the asymptotic estimation result from Olver (1997). Let $n \geq m$, $a > 0$, and $b > 0$

$$\begin{aligned} \frac{a}{bh^m + \mathcal{O}(h^n)} &= \frac{ah^{-m}}{b}\big(1 + \mathcal{O}(h^{n-m})\big), \\ \frac{a}{bh^{-n} + \mathcal{O}(h^{-m})} &= \frac{ah^n}{b}\big(1 + \mathcal{O}(h^{n-m})\big). \end{aligned} \tag{1.109}$$

For the smooth stencil $\mathcal{S}_l \in \text{Smooth}$

$$\begin{aligned} \hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m} &= \frac{\omega_l}{((C+\epsilon)h_0^2 + \mathcal{O}(h^3))^m} \\ &= \omega_l \Big(\frac{h_0^{-2}}{C+\epsilon}(1 + \mathcal{O}(h)) \Big)^m \\ &= \frac{\omega_l}{(D+\epsilon)^m} h_0^{-2m}(1 + \mathcal{O}(h)). \end{aligned} \tag{1.110}$$

In conclusion, we estimate the nonlinear scaling as

$$\hat{\omega}_l = \begin{cases} \dfrac{\omega_l}{(C+\epsilon)^m} h_0^{-2m}(1 + \mathcal{O}(h)), & \text{if} \quad l \in \text{Smooth}, \\ \Theta(1), & \text{if} \quad l \in \text{Jump}. \end{cases} \tag{1.111}$$

Then we estimate the error for the normalized nonlinear weighting (1.105). We expand the sum of nonlinear scaling as

$$\sum_l \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l + \sum_{l \in \text{Jump}} \hat{\omega}_l. \tag{1.112}$$

We estimate the smooth part of the nonlinear weighting as

$$\sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \frac{\omega_l}{(C+\epsilon)^m} h_0^{-2(ar+\eta)}(1 + \mathcal{O}(h)). \tag{1.113}$$

Assume $h < 1$, we keep the largest $ar + \eta(r)$ number as our estimator

$$\begin{aligned} \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l = & \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r_{\max}, r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max} + \eta(r_{\max})}} h_0^{-2(ar_{\max} + \eta(r_{\max}))}(1 + \mathcal{O}(h)) \\ & + \Theta\big(h^{-2(ar + \eta(r))}\big) \\ = & \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r_{\max}, r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max} + \eta(r_{\max})}} h_0^{-2(ar_{\max} + \eta(r_{\max}))} \\ & + \mathcal{O}\big(h^{1-2(ar_{\max} + \eta(r_{\max}))}\big). \end{aligned} \tag{1.114}$$

The overall accuracy estimation of nonlinear weighting (1.105) would be

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}} = \tilde{C} h_0^{2(ar_{\max}+\eta(r_{\max}))}(1 + \mathcal{O}(h)). \tag{1.115}$$

Now we can estimate the reconstruction error as

$$\begin{aligned}
|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| &= |\sum_l \tilde{\omega}_l (u(\mathbf{x} - Q_l(\mathbf{x})))| \\
&\le \sum_{l\in\text{Smooth}} \tilde{\omega}_l |u(\mathbf{x}) - Q_l(\mathbf{x})| + \sum_{l\in\text{Jump}} \tilde{\omega}_l |u(\mathbf{x}) - Q_l(\mathbf{x})| \\
&\le \sum_r \sum_{\substack{l\in \\ \text{Smooth}\cap\mathcal{L}_{r,r}}} \Theta(h^{2[a(r_{\max}-r)+(\eta(r_{\max})-\eta(r))]})\mathcal{O}(h^r) \qquad (1.116) \\
&\quad + \sum_{l\in\text{Jump}} \Theta(h^{2(ar_{\max}+\eta(r_{\max}))})\mathcal{O}(1) \\
&= \mathcal{O}(h^{r_{\max}})
\end{aligned}$$

Combining everything above and (1.102), we arrive at the formal error estimate of the reconstruction.

Lemma 1.7. *Let $\mathcal{R}(\mathbf{x})$ be the reconstruction. Assuming a qusi-uniform mesh, $h_0 = \Theta(h)$, there exists a constant $C > 0$ such that for any $\mathbf{x} \in E_0$,*

$$|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| \le Ch^{r_{\max}}, \tag{1.117}$$

where $r_{\max}$ is defined by (1.107).

1.7.5 Handling Boundary Conditions

It is usually difficult to manage boundary condition for higher order finite volume and finite difference method, which require wide stencils. Some deal with the difficulty use ghost cells or points outside the boundary, and maintain the same stencil as if they were in the interior of the domain. However, the extrapolation or estimation of the manufactured values of ghost cells or ghost points are not straightforward. A notable method, the inverse Lax-Wendroff procedure, substitute the normal derivative of the solution with tangential and time derivatives using the PDE relation in an iteratvie way to impose more precise values for the ghost cells or points.

For assigning boundary condition to a FVM discretization, we should determine whether the flow is inflow or outflow first. If $\mathbf{f}(u_B) \cdot \boldsymbol{\nu} > 0$, we recognize that it is a outflow boundary, therefore, no boundary condition should be assigned. We use "free outflow" boundary that the numerical flux is simply $\mathbf{f}(u_B) \cdot \boldsymbol{\nu}$. The inflow boundary can be understood in different ways. We can understand it as prescribing the inflow flux or fixing the boundary value u_B as an equivelent to Dirichlet boundary in the context of Finite Element.

The key to the above strategy is to have accurate and stable reconstruction near the boundary. ML-WENO gives the flexibility to add stenncils of the desired size that span back into the domain.

1.8 Time Stepping

1.9 Numerical Test

For completeness, we test our ML-WENO algorithm with simple Riemann shock and rarefraction. Let $\mathbf{f}(u) = (u^2/2, u^2/2)$ be the Burgers flux function. In this example, we take the initial condition $u_0(x, y) = 1$, and the computational domain $\Omega = [0, 3] \times [0, 1]$. The true solution is

$$u(x, y, t) = \begin{cases} 0, & x < 0.5, x \geq 0.5t + 1.5, \\ (x - 0.5)/t, & 0.5 \leq x < t + 0.5, \\ 1, & t + 0.5 \leq x < 0.5t + 1.5. \end{cases} \tag{1.118}$$

We forward martching to the time = 2.0, when trailing rarefaction reaches the leading shock.

1.10 Application to Mantle Simulation

Now we discretize equations (**??**), with Finite volume scheme.

1.10.1 Discretization of Enthalpy Transport Equation

1.10.2 Discretization of Composition Transport Equation

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